

ANSWER KEY — Integration Lesson 4 Test: The Antiderivative — Undoing the Derivative

For the grader · full working shown · give credit for a correct function even if $+ C$ is written as a specific number where a family was asked — but note it

1. Find the antiderivative of $8x$. (Give the whole family of answers.)

Work: ask "whose derivative is $8x$?" The derivative of x^2 is $2x$, so $4x^2$ gives $4 \cdot 2x = 8x$. Check: derivative of $4x^2 + C = 8x$ ✓. **Answer:** $4x^2 + C$.

2. Find the antiderivative of 6 (just the constant 6).

Work: a function whose slope is 6 everywhere is the straight line $6x$. Check: derivative of $6x + C = 6$ ✓. **Answer:** $6x + C$.

3. Find the antiderivative of $5x^4$, then check your answer by differentiating it.

Work: the derivative of x^5 is $5x^4$ (the 5 hops down, the power drops to 4) — exactly what we want. Check shown: derivative of $x^5 + C = 5x^4 + 0 = 5x^4$ ✓. **Answer:** $x^5 + C$. (The check must appear for full credit.)

4. Find the antiderivative of $4x + 6$.

Work: term by term. $4x$ comes from $2x^2$ (derivative of $2x^2 = 4x$); 6 comes from $6x$. One $+ C$ covers the sum. Check: derivative of $2x^2 + 6x + C = 4x + 6$ ✓. **Answer:** $2x^2 + 6x + C$.

5. Check whether $x^3 + 2x$ is an antiderivative of $3x^2 + 2$.

Work: differentiate the candidate: $x^3 \rightarrow 3x^2$, and $2x \rightarrow 2$. Total = $3x^2 + 2$, which matches the target exactly. **Answer:** yes — its derivative is $3x^2 + 2$, so it is an antiderivative.

6. Check whether $x^2 + x$ is an antiderivative of $2x$. If not, say what goes wrong.

Work: differentiate the candidate: $x^2 \rightarrow 2x$, and $x \rightarrow 1$. Total = $2x + 1$, but the target is $2x$. The extra $+ 1$ comes from the x term, which shouldn't be there. **Answer:** no — its derivative is $2x + 1$, not $2x$. (Correct antiderivative of $2x$: $x^2 + C$.)

7. Student: "the antiderivative of $6x$ is $3x^2 + 100$." Classmate: "wrong, the answer is $3x^2 + C$." Who is right?

Work: derivative check on the student's answer: derivative of $3x^2 + 100 = 6x + 0 = 6x$ ✓ — so $3x^2 + 100$ genuinely IS an antiderivative of $6x$; it's the family member with $C = 100$. The classmate's " $3x^2 + C$ " is the fuller answer (it names ALL antiderivatives), but calling the student wrong is incorrect. **Answer: the student's answer is valid ($C = 100$); $+ C$ just describes the whole family — both expressions are consistent.**

8. Why does every antiderivative answer end in " $+ C$ "?

Work: the derivative of any constant is 0 (a flat line has no steepness), so differentiating erases added constants. Reversing the process, we can't tell which constant the original function had — x^2 , $x^2 + 7$, $x^2 - 5$ all have derivative $2x$. **Answer: because constants differentiate to 0, infinitely many functions differing by a constant share the same derivative; $+ C$ records that whole family of answers.**

9. $f'(x) = 2x$ and $f(2) = 7$. Find the exact $f(x)$.

Work: antiderivative family: $f(x) = x^2 + C$. Plug in the point: $f(2) = 4 + C = 7 \rightarrow C = 3$. Check: derivative of $x^2 + 3 = 2x$ ✓; $f(2) = 4 + 3 = 7$ ✓. **Answer: $f(x) = x^2 + 3$.**

10. $f'(x) = 6x + 1$ and $f(1) = 9$. Find $f(x)$ and verify both facts.

Work: antiderivative: $6x$ comes from $3x^2$, 1 comes from $x \rightarrow f(x) = 3x^2 + x + C$. Point: $f(1) = 3 + 1 + C = 4 + C = 9 \rightarrow C = 5$. Verify: derivative of $3x^2 + x + 5 = 6x + 1$ ✓; $f(1) = 3 + 1 + 5 = 9$ ✓. **Answer: $f(x) = 3x^2 + x + 5$.**